

SAINT ANDREW'S JUNIOR COLLEGE

Preliminary Examination

MATHEMATICS
Higher 1

8864

Friday

30 August 2013

3 hr

Additional materials : Answer paper
Formula Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.

Answer **all** the questions. Total marks : 95

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically state otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematic steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

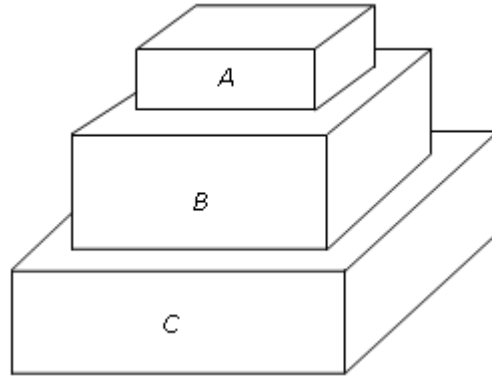
The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This document consists of 6 printed pages including this page.

[Turn Over]

Section A: Pure Mathematics [35 marks]

1



Three rectangular boxes A , B and C are stacked up as shown in the diagram. The total height of the stack is 140 cm. The height of box A is 20 cm while the height of box B and box C are x cm and y cm respectively.

Given $x > y > 20$, and the square of the difference between the height of box B and box C is three times the difference between the height of box A and box B , find x and y . [4]

2

(i) Find the range of values of k for which $kx^2 + 42x + 9k$ is positive for all real values of x . [4]

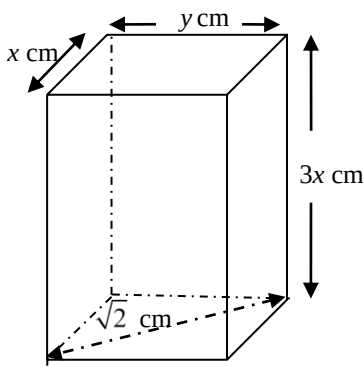
(ii) Hence, find the range of values of x for which $\frac{10x^2 + 132x}{x-1} < 90$. [3]

3

(i) Sketch, on the same axes, the graphs of $y = e^{-x-1} + 2$ and $y = x^3 + x^2 - 3x + 1$, stating the equations of any asymptotes and coordinates of any points of intersection between the curves. [4]

(ii) Find the area enclosed by the two curves, giving your answer to 2 decimal places. [3]

- 4 A rectangular block, x cm by y cm by $3x$ cm as shown in the diagram. The diagonal of the base is $\sqrt{2}$ cm.



- (i) Show that the volume, V , is given by $3\sqrt{(2x^4 - x^6)} \text{ cm}^3$. [2]
- (ii) Without using a graphic calculator, determine the exact value of x for which the volume is maximum. Hence, find the exact maximum value of V . [5]
- 5 A curve C has equation $y = \frac{1-4x}{2x-1}$.
- (i) Sketch C , showing clearly any asymptotes and the coordinates of all points of intersection with the axes. [3]
- (ii) Find the numerical value of the gradient of the tangent to C at the point A where $y = -2.5$. [3]
- (iii) The normal at A meets the x -axis at P , while the tangent at A meets the y -axis at Q . Find the exact area of triangle OPQ . [4]

Section B: Statistics [60 marks]

- 6 70% of girls have a doll while 3 out of 5 girls who own a pet also have a doll.
- (i) Determine whether the events “a randomly chosen girl has a doll” and “a randomly chosen girl owns a pet” are independent, justifying your answer. [1]
- (ii) Let p denote the probability that a girl owns a pet. Find, in terms of p , an expression for the probability that a randomly chosen girl
- (a) has both a doll and a pet [1]
- (b) has either a doll or a pet (or both). [2]
- Hence, find the maximum possible value for p . [1]

- 7 The Principal would like to close down the school gym in order to cut operational costs. The school records indicate that there are 2000 students based on the following breakdown:

	Year 1	Year 2
Arts	400	200
Science	800	600

In order to find out her students' frequency of using the gym, she selects the first 50 students who enter the gym on a particular day.

- (i) Explain why the sampling method above does not produce a random sample. [1]
 - (ii) Describe, in this context, a more representative way to select the sample of students. [3]

- 8 NokiSAPhone is a newly launched popular model of mobile phone. Jacko decided to give himself three attempts to purchase this phone at a nearby shop. The probability that he is successful in his first attempt is p . Subsequently, the success rate is 0.1 more than the success rate of the previous attempt.
 - (i) Draw a tree diagram to represent the possible outcomes. [2]
 - (ii) The probability that he is also unsuccessful in the second attempt given that he is unsuccessful in the first is 0.6. Find the value of p . [1]
 - (iii) Find the probability that he is successful in buying the mobile phone. [2]

- 9 In Saint College, it is known that probability that a student wear glasses is 0.45.
 - (i) A class of 20 students is randomly chosen. Find the probability that more than 15 students wear glasses. [2]
 - (ii) Three classes of 20 students are randomly chosen. Find the probability
 - (a) more than 20 students wear glasses in total, using a suitable approximation. [4]
 - (b) more than 5 students wear glasses in two of the classes and more than 10 students wear glasses in the other class. [3]

Explain why the probability calculated in (a) is more than the probability calculated in (b). [1]

- 10** Land Transport Authority advised drivers to travel 3 seconds behind the car in front of them. This provides the driver with a higher chance of avoiding a collision if the car in front has to brake quickly or is itself involved in an accident. A test was carried out to find out how long it would take a driver to bring a car to rest from the time a red light was flashed. The following results are for one driver in the same car under the same test conditions:

Speed (v km / h)	30	40	50	60	70	80	90
Stopping Time (t s)	1.72	2.38	2.66	2.65	3.12	k	3.83

- (i) It is known that the regression line of t on v is given by $t = 0.84 + 0.033786v$. Find the value of k , correct to 2 places of decimal. [4]
 - (ii) Sketch a scatter diagram for the data. [2]
 - (iii) Find the linear product moment correlation coefficient between v and t and comment on the relationship. [2]
 - (iv) Using a suitable linear model, estimate
 - (a) the stopping time for a speed of 55 km / h,
 - (b) the speed he is driving if the stopping time is 4s. [3]
 - (v) Comment on the reliability of your results in (iv). [2]
- 11** A random sample of 60 bales of hay is delivered by The Hey Company to a farmer. Upon measurement, these turn out to have a mean mass of 44.11kg with a standard deviation of 3kg.
- (i) Find unbiased estimates of the population mean and variance. [2]
 - (ii) If he receives another sample of 50 bales of hay from The Hey Company, estimate the probability that the mean mass of this sample is more than 44.5kg. [3]
 - (iii) The farmer is certain that The Hey Company produces bales that weigh less than 45kg on average. Basing only on the original sample of 60 bales, test at the 5% level of significance whether his belief is valid. [4]
 - (iv) Find the greatest value, correct to 1 decimal place, the mean mass of this sample should be for the farmer's belief to be accepted at the 5% level of significance. [2]

- 12** During the Learning Festival, Jessie makes packets of biscuits with mean mass of 40 g and standard deviation of 2.5 g. She then packs them into cartons containing 55 packets of biscuits in each carton.

- (i) State the approximate distribution of the total mass, S , of the 55 packets of biscuits in a carton and find the probability that S is more than 2230 g. [3]

Assuming that the mass of an empty carton is negligible, find the probability that

- (ii) in a batch of 50 randomly inspected cartons of biscuits, less than 7 weigh more than 2230 g, [2]

- (iii) the total mass of three randomly selected cartons of biscuits exceeds three times the mass of a randomly selected carton of biscuits by more than 10 g. [3]

The label on a carton of biscuits reads 'Net weight 2500g'. A quality regulation requires that at least 90% of such cartons to have a total mass of not less than 2500 g.

- (iv) Assuming that the standard deviation remains unchanged, find the least integer value of the mean mass of a packet of biscuits that would be consistent with this requirement. [4]

~~~~~The End~~~~~